BHARTI AIRTEL LIMITED

DCF Valuation Report

Assumptions, WACC Derivation & Scenario Analysis

April 2026 | *Masters Presentation*

1. WACC Derivation

1.1 Cost of Equity (Ke) = 13.1%

We use the Capital Asset Pricing Model (CAPM): Ke = Rf + βL × (ERP + CRP)

Risk-Free Rate: 6.9%

We use the India 10-year government security yield as the risk-free rate. As of March 2026, this stood at approximately 6.9%, having risen to a 16-month high due to elevated crude oil prices, heavy government borrowing, and fiscal pressures. We use the local-currency risk-free rate because Airtel's cash flows are predominantly in INR. This is sourced from the Reserve Bank of India's G-Sec auction data and Trading Economics.

Equity Risk Premium (ERP): 6.9%

We use Professor Aswath Damodaran's India equity risk premium estimate from his January 2026 update. India's ERP is higher than the US (~5.0%) because of greater macroeconomic volatility, currency risk, regulatory uncertainty, and geopolitical factors. Damodaran's estimate for India typically falls in the 6.5–7.5% range. We use 6.9% as the midpoint. Source: pages.stern.nyu.edu/~adamodar/

Country Risk Premium (CRP): 0.5%

Approximately 28% of Airtel's consolidated revenue (Rs 414 Bn in FY25) comes from 14 African countries including Nigeria, Kenya, Tanzania, Uganda, and others. These countries carry individual sovereign credit risk premiums of 3–6%. We apply a revenue-weighted blended incremental CRP of 0.5% on top of the Indian base ERP. This is conservative — some analysts would use 1.0–1.5%.

Unlevered Beta (βU): 0.73

We use the bottom-up unlevered beta approach, which is more reliable than regression-based betas for companies with volatile stock histories. From Damodaran's January 2026 dataset, the global Telecom (Wireless) sector has an average unlevered beta of 0.73 based on 13 firms. This is low because telecom is a defensive, utility-like sector with relatively predictable cash flows. The sub-1.0 beta reflects that telecom revenues are largely non-cyclical — people do not cancel phone subscriptions in recessions.

Re-Levering for Airtel's Capital Structure

We re-lever the beta using Airtel's market-value D/E ratio. Total debt (FY25, Capital IQ): Rs 2,136,421 Mn. Market capitalisation (March 2026): approximately Rs 10,900,000 Mn. This gives a market D/E of 19.6%. Using the Hamada equation with a normalised tax rate of 25.17%:

βL = 0.73 × (1 + (1 – 0.2517) × 0.196) = 0.73 × 1.147 = 0.84

Note: In the earlier iteration of the model, we mistakenly used a book-value D/E of 0.80, which inflated the re-levered beta to 1.17 and the WACC to 14.1%. Using the correct market-value D/E of 0.196 produces a more accurate and lower beta of 0.84.

Cost of Equity: Ke = 6.9% + 0.84 × (6.9% + 0.5%) = 6.9% + 6.2% = 13.1%

1.2 Cost of Debt (Kd) = 8.5% Pre-Tax, 6.4% After-Tax

Airtel's total debt of Rs 2.14 Tn comprises a mix of INR-denominated bonds (8.0–9.0% coupon), USD bonds (~5.0–6.0%), and lease liabilities. The weighted average pre-tax cost of debt is estimated at 8.5%. The interest coverage ratio has improved from 1.28x (FY21) to 2.70x (FY25) per Capital IQ, reflecting significant de-leveraging.

After-Tax Kd = 8.5% × (1 – 25.17%) = 6.4%

1.3 Capital Structure Weights

Using market values: Market Cap Rs 10.9 Tn and Total Debt Rs 2.14 Tn gives Weight of Equity (We) = 83.6% and Weight of Debt (Wd) = 16.4%. We use market weights rather than book weights because market values better reflect the opportunity cost of capital for investors.

1.4 WACC = 12.0%

WACC = 83.6% × 13.1% + 16.4% × 6.4% = 10.95% + 1.04% = 11.99% ≈ 12.0%

Benchmark	Range	Source
India 10Y G-Sec	6.8–7.0%	RBI
Typical India Large-Cap WACC	10–14%	Market consensus
Airtel WACC (this model)	12.0%	Computed
Jio (estimated)	11–13%	Analyst estimates
US Telecoms (AT&T/T-Mobile)	7–9%	Damodaran

The 12.0% WACC is squarely within the normal range for a high-quality Indian large-cap company. For sensitivity, we test ±100 bps in the bear/bull scenarios.

2. DCF Assumptions

2.1 Revenue Growth

Period	Base Case	Rationale
Year 1–3 (FY26–28)	11–12%	July 2024 tariff hike anniversary effect, expected FY26/27 tariff hike adding Rs 15–20 to ARPU, Homes broadband scaling from 13Mn to 20Mn+ subscribers, Africa growing 20–25% in constant currency
Year 4–7 (FY29–32)	7–9%	ARPU growth moderates as base effect fades, data consumption secular uptrend (30→50+ GB/month), Africa matures to consistent double-digit grower, Enterprise/Airtel Business at 8–10%
Year 8–10 (FY33–35)	6–7%	Converges toward India nominal GDP growth (~6–7%), mature market growth with population + inflation, Africa growth moderates to mid-single digits

The 10-year base case revenue CAGR works out to approximately 8.8%, taking revenue from Rs 1.73 Tn (FY25) to approximately Rs 4.0 Tn (FY35). This is conservative relative to the 5-year historical CAGR of 14.5% (FY21–FY25).

2.2 EBITDA Margins

We project EBITDA margins expanding from 53.9% (FY25) to 57.5% (terminal). Key drivers:

Operating leverage: Telecom is a high fixed-cost business. Each incremental rupee of ARPU drops near-100% to EBITDA since the network cost of serving an existing subscriber is negligible. With 368 Mn mobile subscribers, even Re 1 of ARPU increase generates Rs 4.4 Bn/year at near-full margin.

5G capex cycle maturing: The heavy 5G spectrum and rollout spending of FY23–FY25 is largely complete. As this normalises, amortisation charges moderate. Airtel prepaid Rs 25,980 crores in FY25 to clear deferred spectrum liabilities.

Africa margin expansion: Africa EBITDA margins have risen from 39% (FY21) to nearly 50% as data monetisation deepens and Airtel Money scales. We model a further expansion to 52–54% over the forecast.

Indus Towers effect: Consolidated since November 2024, Indus carries ~57% margins vs Airtel India core at ~59%. This slightly dilutes the blended margin but stabilises over time as tower sharing improves.

2.3 Depreciation & Amortisation

D&A as a percentage of revenue was 26.3% in FY25, elevated due to the 5G rollout and Indus consolidation. We model this declining to 19% by Year 6 as: (a) 5G capex cycle ends, reducing future depreciation charges; (b) spectrum amortisation stabilises post the heavy FY23–FY25 acquisition cycle; (c) revenue growth outpaces asset growth, naturally reducing D&A intensity.

2.4 Capital Expenditure

Capex/revenue was 35.1% in FY25, reflecting the tail end of the 5G build-out. We project this declining to 17% by Year 7 and stabilising there. This is consistent with global telecom norms: mature telecoms spend 15–20% of revenue on capex. The declining path reflects the shift from 5G coverage expansion (capital-intensive) to densification and maintenance (less capital-intensive).

2.5 Tax Rate

We use a normalised tax rate of 25.17% (Indian statutory corporate tax rate) throughout the projection. FY25's effective tax rate was exceptionally low at 2.4% due to large deferred tax credits from the Indus Towers acquisition. We normalise this to avoid distorting the DCF.

2.6 Working Capital

We model the change in net working capital as 1% of the change in revenue. Airtel has consistently negative working capital (customers pay in advance via prepaid), so incremental working capital needs are minimal. The 1% assumption is conservative.

2.7 Terminal Value

Terminal growth rate of 4.5% reflects India's long-term nominal GDP growth (~6–7%) discounted for the fact that Airtel's addressable market will eventually saturate. The terminal value accounts for approximately 57% of the total enterprise value in the base case, which is within the normal range for a DCF (typically 50–75%).

3. Scenario Analysis

3.1 Three-Scenario Summary

	Bear	Base	Bull
WACC	13.0%	12.0%	11.0%
Terminal Growth	3.5%	4.5%	5.5%
10Y Rev CAGR	~5.6%	~8.8%	~10.2%
Terminal EBITDA Margin	50%	57.5%	60%
Terminal Capex/Rev	20%	17%	14%
Implied Price (Rs)	420	1,291	2,459
CMP (Rs)	1,834	1,834	1,834
Upside/(Downside)	-77%	-30%	+34%

Bear Case (Rs 420): Competitive Disruption

Jio's anticipated IPO (2026–27) could incentivise aggressive pricing to protect/grow market share. Vodafone Idea receives government support and becomes a viable third competitor. ARPU growth stalls at Rs 280–290. Starlink or BSNL 4G gains traction in rural areas. African currencies (Naira, Shilling) depreciate sharply, eroding INR-translated revenue. Regulatory risk escalates with higher spectrum fees or extended AGR liabilities. EBITDA margins compress to 50% due to competitive intensity.

Base Case (Rs 1,291): Rational Oligopoly Continues

The three-player oligopoly remains stable. One to two tariff hikes over three years push ARPU toward Rs 300–320 by FY28. Homes broadband crosses 20 Mn subscribers. Africa maintains 15–20% constant-currency growth. 5G capex cycle winds down, with capex/revenue dropping to 17%. Indus Towers integration generates synergies. The Starlink partnership adds incremental revenue. EBITDA margins expand to 57.5%.

Bull Case (Rs 2,459): Premium Growth

Multiple aggressive tariff hikes close the gap with global ARPU benchmarks, pushing ARPU beyond Rs 350. Homes broadband becomes a Rs 200+ Bn business. Africa achieves a profitability breakout with Airtel Money potentially spinning off or IPO-ing at premium fintech valuations. Enterprise digital services (cloud, AI, cybersecurity) grow 20%+ as India's digital transformation accelerates. 5G enterprise use cases generate meaningful B2B revenue. Margins expand above 60%.

3.2 Interpretation

At a CMP of Rs 1,834, the stock is priced between the base and bull case. The implied FY25 EV/EBITDA of ~13.8x is a premium to the 5-year average of 11.5x. This suggests the market is pricing in an optimistic trajectory. The DCF base case implies approximately 30% downside, suggesting the current price already bakes in significant ARPU expansion, margin improvement,

and Africa growth. For the stock to be fairly valued at CMP, you would need assumptions closer to the bull case: sustained 10%+ revenue growth and margins expanding above 58%.

4. Model Structure & Colour Coding

Blue text: Hardcoded inputs (assumptions you can change)

Black text: Formulas that auto-calculate from inputs

Green text: Cross-sheet references (linking DCF Model to Assumptions sheet)

Yellow cells: Key assumption cells — change these to run your own scenarios

Light yellow cells: Key output cells (FCFF, EV, implied price)

The model has 185 formulas and zero errors. Every downstream cell recalculates automatically when you change any yellow input. The WACC on the DCF sheet links directly to the Assumptions & WACC sheet, so changing any WACC input flows through to the implied share price.

5. Data Sources

1. Capital IQ / Refinitiv — CF-Export-02-04-2026.xlsx (consolidated standardised financials, FY21–FY25)

2. Bharti Airtel Annual Reports — FY20 through FY25 (consolidated statements)

3. Bharti Airtel Quarterly KPI Report — 5 quarters Q3FY24 to Q3FY25

4. Damodaran Online — Betas by Sector (January 2026), Equity Risk Premiums, Country Risk Premiums

5. Reserve Bank of India / Trading Economics — India 10Y Government Bond Yield

6. Yahoo Finance / Screener.in — Market capitalisation, share price data

7. TRAI — Telecom subscriber data and performance indicators

8. Airtel Press Releases — 5G rollout, Starlink partnership, spectrum prepayments